PickleScore™ — Wristband v3.0

Assembly Guide — revised August 2026 — body widened to 42×27×15.8mm (was 42×22×12.5mm) after a physical dry-fit found the screw bosses, component stack height, USB-C clearance, and OLED mount all undersized; adapted for JST 1.25mm pigtail connector swap

Bill of Materials

#	Item	Qty	Spec / Notes
1	Seeed XIAO ESP32-C3	1	BLE scanner + built-in LiPo charger
2	OLED display	1	0.91” SSD1306 I2C, 128×32 px
3	LiPo battery	1	301230 (3×12×30mm, ~90mAh) — after JST connector swap
4	JST 1.25mm SH pigtail cable	1	80mm, used to replace factory JST 2.0mm PH on LiPo
5	M1.6 × 5mm screw	4	Pan or flat head — hand driver only
6	20mm quick-release spring bar	2	Standard 20mm lug width
7	20mm silicone sport band	1	Any 20mm quick-release band
8	Hookup wire	~240mm	28–30 AWG — 4 wires × 60mm for I2C harness
9	Heatshrink or tape	small	For insulating connector swap joints

Printed Parts

Part	SHOW setting	Print orientation	Material
Bottom shell	`SHOW = "bottom"`	Curved side down (add brim)	TPU 95A
Top shell	`SHOW = "top"`	Display window face down	TPU 95A

> SCAD file: `ScadFiles/picklscore_wristband_v2.scad` — use `DESIGN = "slim"`

Assembly Steps

0 LiPo Connector Swap

> **Do this first** — the 301230 battery ships with a JST 2.0mm PH connector. The XIAO requires JST 1.25mm SH. The pigtail cable bridges the gap.

1. Cut the JST 2.0mm connector off the LiPo — leave ~20mm of wire on the cell side.
2. Note polarity: **red = +, black = −**. Do not let leads touch.
3. Strip ~3mm on the LiPo leads and the pigtail bare ends.
4. Solder red-to-red, black-to-black. Heatshrink or tape each joint individually before letting them near each other.
5. Verify: plug the new JST 1.25mm end into the ESP32-C3 battery port — XIAO LED should glow. Unplug after confirming.

❶ Flash Firmware

If the XIAO ESP32-C3 was bench-tested previously, firmware may already be loaded. If not, connect via USB-C and upload `PickleScore_Wristband.ino` in Arduino IDE before the board goes into the shell.

❷ Solder I2C Harness (OLED → XIAO)

Cut 4 wires ~60mm. Solder one end to XIAO ESP32-C3 pads, other end to OLED 4-pin header.

XIAO pad	OLED pin
pin 6	SDA
pin 7	SCL
3.3V	VCC
GND	GND

> Check your OLED module's pin label order before soldering — common order is GND / VCC / SCL / SDA left to right, but it varies by vendor.

❸ Bottom Shell — Install XIAO

> **Orientation is rotated 90° from what you might expect:** the board's long edge (the one with the USB-C connector) runs *across the wrist*, not along the arm — the connector needs to reach the port cutout in the side wall, and the integrated strap lugs leave no opening at the arm-direction ends.

1. Drop XIAO ESP32-C3 into the 18×23×5.5mm pocket with its long/USB-C edge running across the wrist, connector pointed at the side-wall port cutout.
2. Confirm USB-C port lines up with the 9×4mm cutout in the side wall.

❹ Bottom Shell — Connect and Place LiPo

1. Plug the swapped JST 1.25mm connector into the XIAO battery port.
2. Lay the 301230 cell flat in the 31×13×3.5mm pocket on top of the XIAO.

❺ Top Shell — Seat OLED

1. Slide the OLED PCB (38×12mm) between the two retaining rails on the underside of the top shell — the rails capture the top/bottom edges (across-wrist direction); the board is already a snug end-to-end fit along the arm direction on its own.
2. The active display area must sit directly over the 24×8mm window cutout.
3. OLED 4-pin header exits toward one short end.

❻ Route Wires and Mate Shells

1. Arrange the 4-wire I2C harness so it loops neatly between shells, clear of all 4 screw boss locations.
2. Lower top shell onto bottom shell. Align the 4 boss holes. Press together.

❼ Drive Screws

Install 4× M1.6×5mm screws. **Hand screwdriver only — stop at snug.** Over-torquing strips TPU bosses.

❽ Attach Strap

1. Insert 20mm quick-release spring bars through the lug channels on each end of the body.
2. Click the silicone sport band ends onto the spring bars.

❾ Functional Test

Power on via USB-C or LiPo. OLED should display:

```
A:00        B:00
Srv:A  #2
```

Once ThroatPod is assembled, press its buttons and confirm the Wristband display updates correctly.

Critical Notes

- **JST polarity:** Always double-check red = + before plugging in. Reversed polarity will damage the XIAO.
- **TPU screws:** M1.6 bosses in TPU strip easily. Never use a powered driver. Stop the moment resistance increases. Screws are still M1.6×5mm — the August 2026 fix corrected the boss height, not the screw length.
- **I2C address:** OLED is at 0x3C. If display stays blank after power-on, check SDA/SCL are not swapped.
- **Wire routing:** If a wire crosses a screw boss it will get pinched when the shells close — check before mating shells.
- **Body size:** now 42×27×15.8mm, up from the original 42×22×12.5mm slim target — a physical dry-fit of the actual XIAO/LiPo/OLED stack (and the USB-C connector's length) needed more room than the original design budgeted. See `picklscore_wristband_v2.scad` header comments for the full before/after breakdown.